

Highlights

Does AI rate through the cycle? Procyclicality in credit assessment

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- First controlled test of procyclicality in LLM credit ratings.
- All five production-tier LLMs downgrade identical firms as the macro worsens.
- Pooled effect: 0.43 rating notches of downgrade per macro-severity step.
- The effect is asymmetric: about 3x steeper into recessions than booms.
- A “through-the-cycle” prompt anchors ratings rather than de-cyclifying them.

Does AI rate through the cycle? Procyclicality in credit assessment

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ABSTRACT

Large language models (LLMs) are increasingly proposed as credit-rating and default-assessment tools, yet whether they rate “through the cycle” or amplify it—a property prudential regulators care about but that has been asserted rather than measured—has remained untested. We hold synthetic firm fundamentals byte-identical across an ordered five-point macroeconomic-severity axis (plus a matched credit-irrelevant placebo axis) and elicit letter ratings and one-year default probabilities from five production-tier models (two families, two capability tiers), so that any rating movement is procyclicality by construction, identified without the unobserved-fundamentals confound that limits the classical Amato–Furfine test. Across 32,000 ratings, every model is significantly procyclical (pooled 0.429 rating notches of downgrade per severity step, 95% CI [0.41, 0.45]; a 1.76-notch boom-to-recession swing), the effect is sharply asymmetric (3.2× steeper into recessions than out of them), an order of magnitude larger than the placebo, and robust to model-version fixed effects and a token-logprobability estimator. An explicit “through-the-cycle” instruction lowers the rating slope but does not restore through-the-cycle behaviour—it anchors the rating while suppressing the point-in-time default probability that should move, whereas a naive “be stable” instruction preserves that distinction better. Because model choice and prompt framing are first-order determinants of the effect, LLM-based credit assessment can silently import the capital-procyclicality channel that regulation seeks to dampen.

1. Introduction

Credit ratings are important for debt pricing, investment decisions, and regulatory capital requirements. A perennial concern is that ratings could be procyclical, with companies downgraded during recessions and upgraded during expansions even when the underlying credit fundamentals have not materially changed. Such rating changes can exacerbate the credit cycle by raising borrowing costs and, through internal-ratings-based (IRB) capital rules, capital requirements when economic conditions are already weak (Kashyap and Stein, 2004; Gordy and Howells, 2006). Rating agencies therefore attempt to capture creditworthiness “through the cycle” (TTC), placing more importance on a borrower’s ability to withstand temporary macroeconomic fluctuations than on mechanically reacting to current conditions (Amato and Furfine, 2004; Löffler, 2004). However, whether these ratings actually serve this purpose remains debated (Löffler, 2013).

There is growing interest in using large language models (LLMs) for financial decision-making, including credit decisioning and risk assessment, and supervisors have flagged that generative models could re-introduce procyclicality (Aldasoro, Gambacorta, Korinek, Shreeti and Stein, 2024). This concern is justified: LLMs behave as context-sensitive economic agents (Horton, 2023; Binz and Schulz, 2023), their outputs are highly sensitive to the way prompts are framed (Tversky and Kahneman, 1981), and a shared foundation model could homogenise decisions across users (Bommasani, Creel, Bansal, Guha and Liang, 2022). Yet the size of this effect on credit ratings has not been directly measured. Identifying procyclicality in observational studies of human agencies is difficult because firm fundamentals and macroeconomic conditions tend to move together, so the classical ordered-probit test of Amato and Furfine (2004) must proxy for unobserved fundamentals. Recent evidence that frontier LLMs reproduce date-specific human economic sentiment (Graham, Harvey and Jha, 2026) makes the question both timely and concrete.

This paper uses a controlled experimental setting to test the procyclicality of LLM-based credit rating. We keep the firm fundamentals unchanged and only change the macroeconomic environment, so any systematic change in the assigned rating can be attributed to the macroeconomic context and not to changes in firm credit quality. The benchmark is not that any movement is bad: a genuine through-the-cycle rater

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should hold the *rating* steady while its point-in-time default probability still tracks conditions, and we test that joint condition directly. We make three contributions. We first show that all five production-tier models tested downgrade identical firms as the state of the economy deteriorates, but the magnitude differs substantially across models. Second, we find a strong asymmetry, with rating sensitivity significantly larger in recessions than in expansions. Third, we test whether this behaviour can be mitigated via prompt design: an explicit through-the-cycle instruction decreases rating movement, but it also suppresses the one-year default probability's responsiveness, suggesting that the model anchors its outputs instead of distinguishing between point-in-time risk and through-the-cycle ratings (Tversky and Kahneman, 1974). Section 2 positions the contribution; Section 3 sets out the design; Section 4 reports the results; Section 5 concludes.

2. Related literature

Our study relates to three strands of literature. The first relates to the assessment of through-the-cycle and rating procyclicality. Amato and Furfine (2004) test the response of credit ratings to the business cycle, controlling for firm fundamentals, and Löffler (2004, 2013) examine the extent to which rating agencies achieve through-the-cycle stability. The prudential stakes are set by the interaction of ratings with Basel capital rules (Kashyap and Stein, 2004; Gordy and Howells, 2006), with sovereign and corporate ratings alike carrying real pricing consequences (Cantor and Packer, 1996).

The second strand is the rapid adoption of *LLMs in finance*, for which this journal has become a leading outlet (Ali, Zafar and Aysan, 2025). Recent applications include return prediction (Lopez-Lira and Tang, 2023), financial-statement analysis (Kim, Muhn and Nikolaev, 2024), financial advice (Fieberg, Hornuf, Meiler and Streich, 2025; Kirschbaum, Keiper, Mölders and Zülch, 2026), and use as simulated economic agents (Horton, 2023); within finance specifically, models have been turned to measuring monetary-policy uncertainty (Ito, Sato and Ota, 2025), assessing the financial conditions of AI firms (Giaretta and Zen, 2026), and—most relevant here—to reliability evidence in banking supervision (García-Llorente and Olmeda, 2026). A parallel line warns that machine-generated scores and disclosures can mislead, from greenwashing in ESG scores (Kathan, Utz, Dorfleitner, Eckberg and Chmel, 2025) to “AI washing” in corporate disclosure (Song, Hou, Ouyang and Hao, 2026), while broader analyses consider AI's systemic and macroeconomic footprint, from agent-based market (in)stability (Llacay and Pfeffer, 2026) to the skill premium (Bloom, Prettnner, Saadaoui and Veruete, 2025). Most closely, Hens and Nordlie (2025) compare the risk profiles LLMs produce for investors with those of experienced bankers, finding relatively small differences in the levels of risk assigned, although the models give weaker explanations. We examine a different dimension of reliability—whether an LLM alters its credit assessment in response to economically irrelevant changes in the macroeconomic context surrounding otherwise identical firm information.

The third strand is the susceptibility of LLM outputs to framing and contextual information. Prior work documents anchoring, framing effects, and imperfect calibration in language-model decisions (Tversky and Kahneman, 1974, 1981; Binz and Schulz, 2023; Kadavath, Conerly, Askell et al., 2022), and shows that shared foundation models can homogenise decisions across users (Bommasani et al., 2022). In our setting these behavioural properties connect to a specific financial mechanism: if the macroeconomic context systematically affects credit ratings even when the underlying firm fundamentals do not change, contextual sensitivity translates directly into rating procyclicality.

3. Design and identification

The unit of analysis is a rating cell = (model, prompt framing, seed, firm, macro state). We construct 80 synthetic firms spanning five credit-quality tiers and eight sectors, and characterise each firm using the same set of credit fundamentals: revenue growth, EBITDA margin, leverage, interest coverage, free cash flow, liquidity, and Altman-style financial ratios (Altman, 1968). The information about each firm presented to the models is the same in all experimental conditions; each firm's fundamentals are serialised once and reused byte-identically under every macro state. Following this, we merely alter the model's exposure to the macroeconomic environment by switching it between five ordered states—boom, expansion, neutral, slowdown, and severe recession (severity $-2 \dots +2$). A common set of macroeconomic indicators is used to represent each state, including GDP growth, unemployment, credit spreads, default rates, and lending

standards. No calendar dates are given, reducing the chance that a model associates a situation with a particular historical event.

To strengthen identification, we add two controls. We first generate a matched placebo condition carrying credit-irrelevant information about regional weather and traffic. The placebo has the same format and a similar amount of numerical information as the macroeconomic scenarios but should not affect a firm's creditworthiness; it helps us differentiate macroeconomic sensitivity from a more general tendency to respond to changing contextual information, and the net (macro minus placebo) coefficient is the demand-effect-adjusted cyclicity. Second, we keep the same random seed for a given firm and prompt framing across all macroeconomic states, so that sampling variation cannot be misinterpreted as a cyclical effect. Each assessment produces a letter credit rating (mapped to an S&P notch on a 21-point scale) and a one-year probability of default (PD).

We consider four prompt framings: a terse baseline, a point-in-time instruction, an explicit through-the-cycle instruction, and a generic "be stable" instruction that makes no reference to the credit cycle. The experiment uses five production-tier models from two families—three from OpenAI (`gpt-4o-mini`, `gpt-4.1-mini`, `gpt-4o`) and two Google Gemini Flash-tier models accessed via the `gemini-flash-latest` and `gemini-flash-lite-latest` aliases—all queried in July 2026 with the served-model snapshot (`system_fingerprint`) logged per call for reproducibility. For each condition each model is run twice at temperature zero. Our main dependent variable is the change in rating notch for a firm as macroeconomic severity increases, with a positive coefficient implying greater procyclicality. We report average firm slopes and 95% confidence intervals based on 2,000 firm-clustered, tier-stratified bootstrap draws. We also look at the total boom-to-recession rating swing, the difference in downside and upside sensitivity, the Amato–Furfine ordered-probit coefficient, and—for the OpenAI models—a continuous expected-notch slope from the rating-token log-probabilities. The whole experiment yields 32,000 credit assessments for US\$13.69, with zero parse errors.

4. Results

4.1. LLMs are procyclical on identical fundamentals

On byte-identical fundamentals, all five models downgrade the same firms as the described macroeconomy worsens (Table 1; Fig. 1a). Pooling across models under the terse baseline, ratings fall by 0.429 notches per severity step (95% CI [0.41, 0.45])—a 1.76-notch swing from boom to severe recession, enough to move a BBB firm toward BB+/BB with nothing real changing. The per-model coefficient ranges from 0.230 for Gemini Flash to 0.775 for GPT-4o-mini, and no model's confidence interval contains zero, so the choice of model is itself a first-order determinant of procyclicality—a roughly threefold spread. Among the OpenAI models, GPT-4o is markedly less sensitive than GPT-4o-mini, with slopes of 0.387 and 0.775 respectively.

The result is robust across several checks, detailed in Appendix A: the matched credit-irrelevant placebo moves ratings only 0.042 notches per step, so the net macro-minus-placebo effect is 0.387 (95% CI [0.36, 0.41]), still excluding zero, and the coefficient survives served-model fixed effects, a rating-token log-probability estimator, and the Amato–Furfine ordered-probit null (Table 3).

The magnitudes are economically material. A pooled 1.76-notch swing means that a firm with unchanged fundamentals—rated, say, A—in a boom—can be rated near BBB in a severe recession purely because of the macroeconomic environment in which the model is placed, and for the most sensitive model the boom-to-recession swing exceeds three notches (§4.4).

4.2. The procyclicality is asymmetric

The response to macroeconomic conditions is highly asymmetric. Splitting the macro axis at neutral, the pooled downside half-slope (neutral to severe recession) is 0.67 notches per step (95% CI [0.63, 0.72]) versus an upside half-slope (boom to neutral) of 0.21 (95% CI [0.18, 0.24]), so downgrade sensitivity is about 3.2 times larger than upgrade sensitivity, a pattern seen in all five models (Fig. 2). This asymmetry suggests that LLM ratings are more sensitive to deteriorating than to improving conditions, and it matters because downgrades in downturns raise borrowing costs and capital requirements exactly when financial conditions are already under strain; it also echoes the stricter-in-downturns behaviour long attributed to human agencies (Löffler,

2004). Because capital requirements are convex in the rating and bind hardest in stress, an asymmetric downgrade profile is more procyclical than a symmetric one of equal average slope.

4.3. “Through-the-cycle” instructions anchor rather than de-cyclify

Prompt framing is a strong determinant of rating sensitivity: the estimated slope declines from 0.516 for the point-in-time instruction to 0.429 for the terse baseline, 0.329 for the “be stable” instruction, and 0.160 for the explicit through-the-cycle instruction (95% CI [0.14, 0.18]; terse-minus-TTC contrast 0.269). At first sight this suggests that through-the-cycle prompting successfully reduces procyclicality. It does not. Genuine through-the-cycle behaviour is a joint condition: the rating slope should fall toward zero *and* the point-in-time default probability should keep moving with the cycle. We measure the default probability on the log-odds scale, so the comparison across framings is not driven by level differences and is robust to the miscalibration of verbalised LLM probabilities (Kadavath et al., 2022; Kang and Liu, 2023). Pooled, the log-odds PD slope falls from 0.270 under the terse prompt to 0.102 under the through-the-cycle instruction—a retention of only 0.38 (95% CI [0.35, 0.40]), far below the value near one that genuine through-the-cycle behaviour requires, and the suppression is comparable on the downside and upside (retention 0.35 and 0.38). The instruction thus stabilises the rating but also suppresses the default probability that should remain sensitive: it anchors the wrong object. By contrast, the vaguer “be stable” instruction, which is not semantically about the cycle, preserves more default-probability movement while still calming the rating. A residual downgrade survives everywhere—up to 1.63 notches across the cycle for GPT-4o-mini.

4.4. Capital consequences

The material question is not merely that the rating moves but what the movement costs. We translate the rating path into capital by anchoring through the *rating*—which is what enters a credit pipeline—rather than the disclaimed verbalised PD: we map each notch to a stylised one-year default probability calibrated to published agency average default rates, and apply the Basel internal-ratings-based (IRB) corporate risk-weight formula as a stylised proxy for the core economic logic (parameters and mapping in Appendix A). Holding fundamentals fixed and reporting the *change* across the severity axis, an identical firm’s required capital rises by a pooled 1.87 percentage points of exposure from boom to severe recession (95% CI [1.73, 2.00]), and by 3.48 pp for the most sensitive model; the verbalised-PD variant agrees in direction (2.92 pp). Because the risk weight is convex in PD, the §4.2 asymmetry compounds: the downside capital increase (neutral→recession, 1.55 pp) is about five times the upside decrease (boom→neutral, 0.32 pp)—a wider gap than the notch asymmetry alone. These findings imply that procyclical LLM ratings can have economically significant downstream consequences even when the firm’s fundamentals do not change at all.

5. Conclusion

Using a design in which fundamentals are fixed by construction, we provide a controlled measurement of rating procyclicality in LLM-based credit assessment. All five models downgrade identical firms as economic conditions are described as worsening, by economically meaningful and statistically robust margins, and they do so asymmetrically—punishing simulated downturns far more than they reward simulated expansions. The magnitude is a first-order function of model identity, so in a deployed pipeline the choice of model is itself a prudential decision.

Prompt design mitigates this behaviour only partially. An explicit instruction to rate through the cycle reduces rating sensitivity, but it also diminishes the responsiveness of the one-year default probability; the models appear to stabilise both outputs rather than separating a through-the-cycle rating from point-in-time default risk. Translating the rating changes into capital requirements produces economically meaningful swings even when the underlying firm fundamentals have not changed.

The synthetic firm battery is intentional. The experiment is built for identification, not to capture the full complexity of real borrowers—yet varying the macroeconomic context around fixed firm information captures an exposure that arises in real-world LLM credit pipelines, where models may access macroeconomic information through market data, news, retrieval systems, or analyst inputs. The results should therefore be read as evidence for a specific behavioural mechanism rather than an evaluation of deployed credit systems. We study a synthetic firm battery, five production-tier models from two families, and prompt-level

Table 1

Per-model rating procyclicality on byte-identical fundamentals. β = notches of downgrade per +1 macro-severity step (terse framing); firm-clustered bootstrap 95% CIs. Swing = boom→severe-recession notches. Asymmetry = downside (into recession) vs upside (into boom) half-slopes. Residual = β after the through-the-cycle instruction; PD-ret. = PD-slope retention under that instruction.

Model	β	95% CI	Swing	Down	Up	D:U	Resid.	PD-ret.
Gemini Flash	0.230	[0.19, 0.27]	0.96	0.38	0.10	4.0	0.059	0.22
Gemini Flash-Lite	0.285	[0.25, 0.32]	1.15	0.45	0.12	3.6	0.056	0.34
GPT-4.1-mini	0.471	[0.41, 0.53]	1.94	0.77	0.20	3.9	0.096	0.24
GPT-4o	0.387	[0.34, 0.43]	1.55	0.59	0.18	3.3	0.187	0.52
GPT-4o-mini	0.775	[0.69, 0.86]	3.18	1.15	0.44	2.6	0.403	0.59

Table 2

Pooled procyclicality by prompt framing (across five models), notches per severity step with firm-clustered bootstrap 95% CIs.

Framing	Pooled β	95% CI	Excl. 0
Point-in-time	0.516	[0.49, 0.54]	True
Terse	0.429	[0.41, 0.45]	True
"Be stable"	0.329	[0.30, 0.35]	True
Through-the-cycle	0.160	[0.14, 0.18]	True

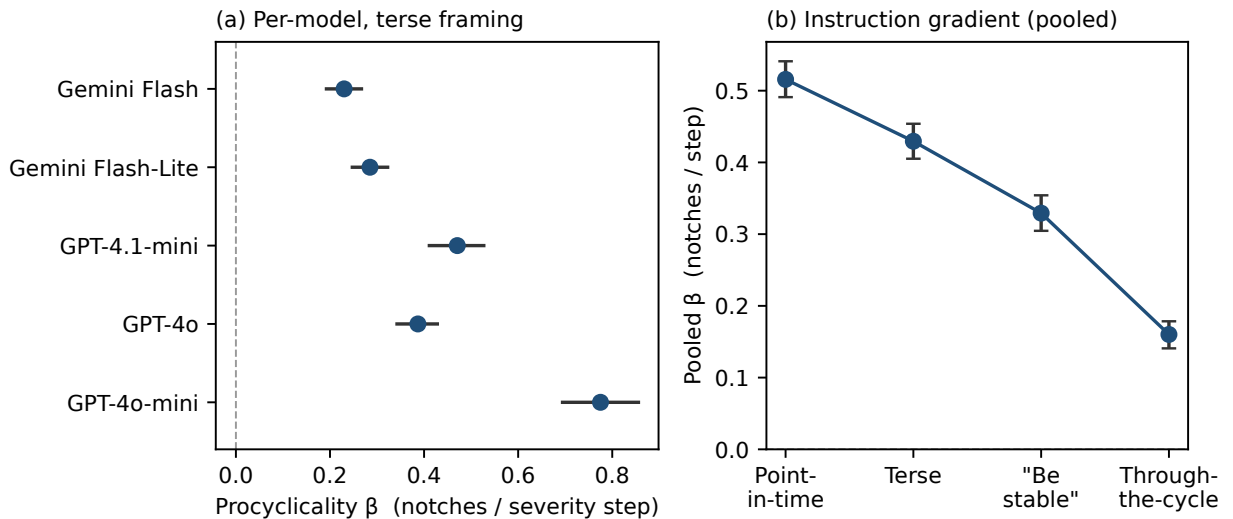


Figure 1: (a) Per-model procyclicality β under the terse framing with 95% CIs; the dashed line marks the through-the-cycle null ($\beta = 0$). (b) Pooled β across the four prompt framings (point-in-time → terse → "be stable" → through-the-cycle) with 95% CIs.

interventions; future work could benchmark LLM behaviour against agency and internal-rating baselines and explore whether fine-tuning, retrieval design, or other model-level interventions can reduce procyclicality without suppressing genuine changes in credit risk.

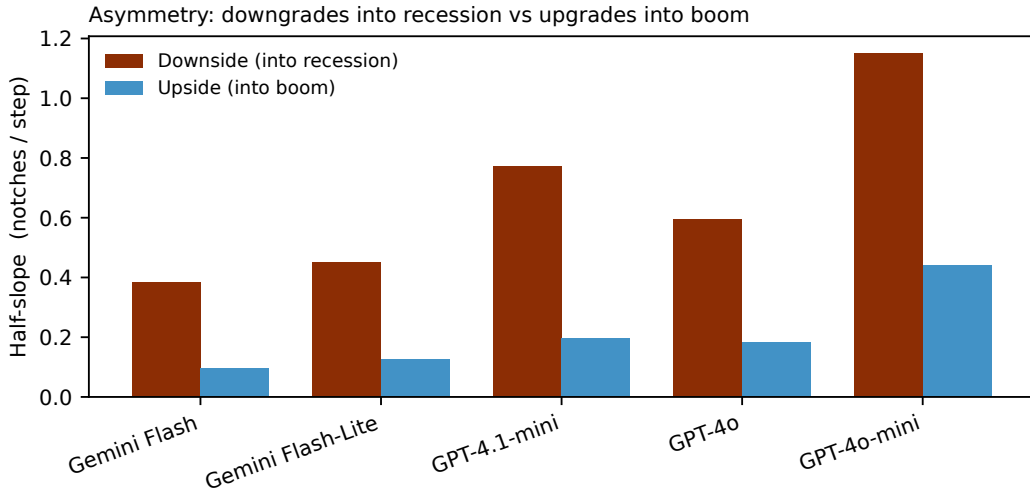


Figure 2: Asymmetry: per-model downside half-slope (neutral→severe recession) versus upside half-slope (boom→neutral), terse framing. Every model downgrades far faster into recessions than it upgrades into booms.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in implementing the data-capture and analysis pipeline and in drafting and editing the manuscript text. The large language models evaluated in this study—three OpenAI models and two Google Gemini models—constitute the object of study of this paper, and their use is fully described in Section 3; they are not tools of manuscript preparation. All AI-assisted content was carefully reviewed, edited, and verified by the authors, who take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data and code required to reproduce every result are openly available in a public repository. The citable archive (DOI) is withheld here to preserve author anonymity for double-blind review and will be provided on acceptance (Anonymous Authors, 2026).

A. Robustness

Placebo. Under the matched credit-irrelevant placebo axis, ratings shift by only 0.042 notches per step, so the net (macro minus placebo) effect is 0.387 notches per step (95% CI [0.36, 0.41]). *Infrastructure.* Adding served-model (`system_fingerprint`) fixed effects shifts the terse coefficient by < 0.003 for every OpenAI model; a within-cell seed noise floor confirms the signal exceeds residual non-determinism. *Parsing.* Every model parses at 1.000, with the most cyclical model (GPT-4o-mini) among the cleanest, so the effect is not a

Table 3

Robustness of the terse-framing coefficient: placebo-adjusted (net), log-probability expected-notch (OpenAI only), within-cell seed noise floor, and the Amato–Furfine ordered-probit estimate.

Model	β	Net	Exp-notch	Noise SD	Probit
Gemini Flash	0.230	0.190		0.020	0.272
Gemini Flash-Lite	0.285	0.268		0.000	0.235
GPT-4.1-mini	0.471	0.464	0.420	0.086	0.307
GPT-4o	0.387	0.367	0.378	0.025	0.403
GPT-4o-mini	0.775	0.646	0.765	0.072	0.726

censoring or parsing artefact. *Log-probabilities.* For the OpenAI models, a continuous expected-notch slope from the rating-token log-probabilities reproduces the sampled estimate (e.g., GPT-4o-mini 0.775 vs 0.765). *Ordered probit.* The Amato–Furfine null uses the firm mean notch as a Mundlak control and agrees in sign throughout, of comparable magnitude, though the two estimators do not perfectly preserve the cross-model rank order (Table 3). *Capital mapping.* The capital translation (§4.4) maps each notch to a stylised one-year default probability (AAA $\approx$ 0.01%, BBB $\approx$ 0.25%, BB $\approx$ 1.1%, B $\approx$ 5%, CCC $\approx$ 20%; monotone across the 21-notch scale, calibrated to published agency averages) and applies the Basel IRB corporate formula (loss given default 0.45, maturity 2.5, the corporate asset-correlation function; no SME or firm-size adjustment); the reported *change* in capital is robust to the exact mapping.

B. Prompts

The prompt framings and the macro and placebo blocks below are extracted programmatically from the capture code (`agent.py`) and reproduce the executed instrument verbatim.

B.1. Prompt framings

The four instruction framings (system prompt), verbatim as executed:

v_terse

You are a credit rating analyst. Assign a long-term issuer credit rating to the company described

v_ttc

You are a credit rating analyst applying a THROUGH-THE-CYCLE methodology. Assign a long-term issuer

v_pit

You are a credit rating analyst applying a POINT-IN-TIME methodology. Assign a long-term issuer

v_stable

You are a credit rating analyst. Assign a long-term issuer credit rating that is STABLE and does

B.2. Macro-environment blocks

Prepended to the byte-identical fundamentals; only these five blocks differ across the severity axis:

boom

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +4.5%
- Unemployment rate: 3.4%
- Investment-grade corporate credit spread: 95 bps
- Trailing 12-month corporate default rate: 0.6%
- Bank lending standards: easing broadly

expansion

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +2.6%
- Unemployment rate: 4.2%
- Investment-grade corporate credit spread: 135 bps
- Trailing 12-month corporate default rate: 1.4%
- Bank lending standards: modestly easing

neutral

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +1.8%
- Unemployment rate: 5.0%
- Investment-grade corporate credit spread: 175 bps
- Trailing 12-month corporate default rate: 2.3%
- Bank lending standards: unchanged

slowdown

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): -0.8%
- Unemployment rate: 6.4%
- Investment-grade corporate credit spread: 320 bps
- Trailing 12-month corporate default rate: 4.1%
- Bank lending standards: tightening

severe_recession

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): -3.6%
- Unemployment rate: 8.9%
- Investment-grade corporate credit spread: 620 bps
- Trailing 12-month corporate default rate: 7.5%
- Bank lending standards: sharply tightening

B.3. Placebo blocks (credit-irrelevant control)

Matched in structure and number density, on a credit-irrelevant axis:

placebo_calm

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.6 degC
- Daylight hours vs annual mean: +4.5%
- Monthly rainfall vs normal: 95 mm
- Regional air-quality index: 38
- Commuter traffic congestion: light

placebo_mild

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.2 degC
- Daylight hours vs annual mean: +2.6%
- Monthly rainfall vs normal: 135 mm
- Regional air-quality index: 52
- Commuter traffic congestion: moderate

placebo_normal

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: 0.0 degC
- Daylight hours vs annual mean: +1.8%
- Monthly rainfall vs normal: 175 mm
- Regional air-quality index: 60
- Commuter traffic congestion: typical

placebo_rough

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -0.4 degC
- Daylight hours vs annual mean: -0.8%
- Monthly rainfall vs normal: 320 mm
- Regional air-quality index: 88
- Commuter traffic congestion: heavy

placebo_storm

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -1.1 degC
- Daylight hours vs annual mean: -3.6%
- Monthly rainfall vs normal: 620 mm
- Regional air-quality index: 140
- Commuter traffic congestion: severe

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